

■ Steel Surface Inspection Report

AI-Powered Defect Detection & Root Cause Analysis

Parameter	Value
Detected Defect	Rolled-In Scale
Confidence	88.84%
Model Confidence	■ High
Affected Area	42.86%
Defect Severity	■ Critical
Risk Score (RPN)	9/10
Risk Level	■ HIGH

Description

Oxide scale folded into the steel surface during rolling.

Probable Root Causes

- Inadequate descaling before hot rolling passes
- Excessive furnace scale formation (high temp / long soak)
- Low descaling water pressure or blocked nozzles

Process Parameters to Check

- Check descaler nozzle condition and pressure logs
- Review reheating furnace temperature and soaking time
- Inspect inter-pass descaling between rolling stands

Recommended Corrective Actions

- Increase descaling pressure to ≥ 180 bar
- Reduce furnace soaking time at peak temperature
- Schedule regular descaler nozzle maintenance

All Class Probabilities

Defect Class	Probability
Crazing	4.99%
Inclusion	1.05%
Patches	2.13%
Pitted Surface	2.26%
Rolled-In Scale	88.84%
Scratches	0.74%